

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 1: Rational and Irrational Numbers

Every formula and result of the chapter on one sheet

|    |                                               |                                                                                |
|----|-----------------------------------------------|--------------------------------------------------------------------------------|
| 1  | Rational number                               | $\frac{p}{q}$ , $p$ and $q$ integers, $q \neq 0$                               |
| 2  | Terminating decimal                           | Denominator (lowest terms) has only the prime factors 2 and 5                  |
| 3  | Rational between $a$ and $b$                  | $\frac{a+b}{2}$                                                                |
| 4  | Irrational number                             | Decimal neither terminates nor recurs                                          |
| 5  | Product rule                                  | $\sqrt{ab} = \sqrt{a}\sqrt{b}$                                                 |
| 6  | Quotient rule                                 | $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$                               |
| 7  | Conjugate product                             | $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$                                       |
| 8  | Conjugate product                             | $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$                           |
| 9  | Square of a sum                               | $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}$                                 |
| 10 | Rationalising $\frac{1}{\sqrt{a}}$            | $\frac{\sqrt{a}}{a}$                                                           |
| 11 | Rationalising $\frac{1}{\sqrt{a} + \sqrt{b}}$ | $\frac{\sqrt{a} - \sqrt{b}}{a - b}$                                            |
| 12 | Useful values                                 | $\sqrt{2} \approx 1.414$ , $\sqrt{3} \approx 1.732$ , $\sqrt{5} \approx 2.236$ |